

ARCHIT AGRAWAL

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EXPERIENCE

AI Software Engineer - EdPlus, Arizona State University

Tempe, AZ | Jun 2026–Present

- Architected and shipped a reusable survey-analysis platform of 10 agents and 65 registered tools; replaced bespoke pipelines with a declarative TypeScript architecture on Bedrock, AgentCore, Step Functions, and CDK.
- Prevented roughly 12% silent row loss through schema-constrained decoding, response correlation tokens, and citation verification; achieved 100% join integrity and zero label churn across repeat runs.
- Made newly uploaded surveys immediately queryable without dataset-specific configuration by building a natural-language semantic layer over DuckDB.
- Redirected the team roadmap through a controlled evaluation showing model choice moved labeling F1 by 0.230 - roughly 33× the 0.007 movement from pipeline architecture.
- Eliminated a Distributed Map race through partitioned writes and fan-in, verified across 258 runs; held query latency flat at 200× row volume with DuckDB over Lance on S3.
- Delivered the platform through a 14-page Nuxt 4 application and 10-module NestJS API compiled against one shared TypeScript contract.
- Built 1,255 tests across 592 modules, covering deterministic transforms, agent tools, repair paths, replay behavior, and end-to-end platform contracts.

Founding AI/ML Engineer - MyStage Music Inc.

Remote | Jul 2025–Present

- As founding AI/ML engineer, cut service dependencies 64% (14 to 5) by re-architecting a 14-Cloud-Function pipeline as one checkpointed LangGraph worker on Cloud Run.
- Indexed 70,000+ records daily across 650+ venues through a Playwright service with proxy rotation and adaptive backoff.
- Improved downstream search accuracy 25% with Gemini entity resolution and deduplication before Firestore ingestion.
- Served search at sub-50 ms latency through FastAPI and Algolia with real-time Firestore synchronization.
- Made concurrent retries safe with atomic Firestore claims and deterministic task IDs; added human-in-the-loop pause and resume.
- Instrumented cross-worker execution with Logfire distributed tracing and wrote 200+ pytest tests covering routing, retries, and failure isolation.

Software Engineer - EdPlus, Arizona State University

Tempe, AZ | Sep 2023–May 2025

- Led the end-to-end development of a multi-tenant RAG assistant used by 1,000+ faculty authoring courses for 60,000+ students.
- Made transcript analysis 16× faster, reducing it from four hours to 15 minutes with a Neo4j knowledge graph and schema-constrained, LLM-generated Cypher.
- Reduced college onboarding from a deployment to a configuration change while enforcing tenant isolation at retrieval rather than prompt level.
- Improved exact-token queries such as course codes with hybrid vector and keyword retrieval; evaluated groundedness, relevance, and coherence.

Data Research Aide - Knowledge Exchange for Resilience, ASU

Tempe, AZ | Jun 2024–Aug 2024

- Reduced p95 retrieval latency 60% at 95% recall with hybrid FAISS and Neo4j retrieval; cut API latency nearly 90% (198 ms to 20 ms) through Redis and SQL optimization.
- Delivered 20+ REST APIs for an NSF-funded platform using .NET 8, Dapper, MediatR, JWT, and SQS.
- Ingested 10,000+ faculty profiles daily through an idempotent ETL pipeline with Postgres quality gates.

Software Engineer - Zeus Learning

Mumbai, India | Jan 2022–Jul 2023

- Reduced resource usage 35% and infrastructure cost 20% by migrating a .NET monolith to Kubernetes microservices through a strangler pattern.
- Reduced deployment time 70% and incidents 40% with SonarQube merge gates and containerized delivery.
- Shipped Redis-backed demand prediction across 300+ enterprise sites, including Goldman Sachs and Merck; reported occupancy increased 30%.

Product Intern - EAT.FIT

Bengaluru, India | Sep 2021–Dec 2021

- Reduced driver-location serving cost 45% through distance-aware polling and client interpolation.
- Built a backpressure-aware WebSocket notification service with idempotent redelivery.

EDUCATION

M.S., Computer Science, Arizona State University - GPA: 4.0

Aug 2023–May 2025

B.Tech., Information and Communication Technology, DA-IICT

Aug 2018–May 2022

SKILLS

Languages: Python, TypeScript, C#, SQL, JavaScript

AI & Data: LangGraph, MCP, Bedrock/AgentCore, Vertex AI, OpenAI, RAG, DuckDB, PostgreSQL/pgvector, Neo4j, FAISS, Lance, Algolia, Redis, Firestore

Backend & Cloud: FastAPI, NestJS, .NET 8, React, Nuxt 4, AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions

Quality: pytest, Vitest, Locust, Logfire, distributed tracing